

Help Learn More About a Unique Solar Eclipse Phenomenon

Join the 2026 Eclipse Shadow Band Citizen Science Project

Experience Something Rare

During the few moments before and after totality, rippling patterns called shadow bands race across the ground. This citizen science project invites eclipse observers across Iceland and Spain to record shadow bands and help answer questions that remain largely unexplored.

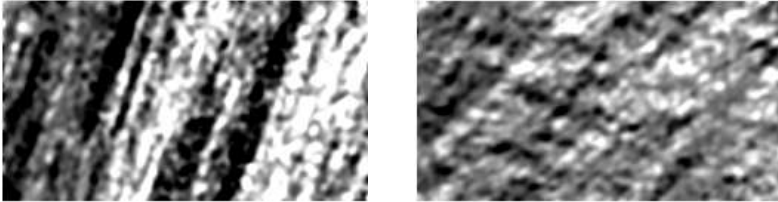


Figure 1: Eclipse shadow bands extracted from a video recording through image processing. Bands on the right image are multi-modal (two sets of bands traveling in opposite directions simultaneously).

Why Participate?

- Witness one of nature's rarest optical phenomena.
- Contribute to genuine scientific research.
- Help answer questions about how location and altitude influence shadow bands.
- Be part of an international citizen science effort.
- Receive detailed observing instructions before the eclipse.

What You'll Do

- Build a simple observation board using inexpensive materials (cardboard boxes).
- Record shadow bands with your smartphone or camera during the eclipse.
- Submit your video and observation details after the event.
- The observation itself takes only a few minutes during totality.

We're Looking for Volunteers In

- Locations near the eclipse centerline in Iceland or Spain.
- Locations near the edge of totality in Iceland or Spain.
- High-altitude observation sites in Iceland or Spain.
- If you're already planning to travel to Iceland or Spain for the eclipse, your observations could be especially valuable.

Materials Needed

- Smartphone or digital camera with video capability
- Tripod or phone stand
- White bed sheet or four white poster boards
- Five cardboard boxes
- Packing tape

Safety First

Never look directly at the Sun without certified eclipse glasses or other approved solar viewing equipment.

Join the Project

To register, email: jconti@alum.mit.edu. Registered volunteers will receive detailed instructions for building the observation board and recording data.

About the Organizer

Joe first witnessed eclipse shadow bands during the April 2024 North American total solar eclipse. Fascinated by this little-understood phenomenon, he developed image-processing techniques to analyze recorded shadow bands and organized this independent citizen science project to gather observations from multiple locations during the 2026 eclipse. See the following preprint paper: <https://arxiv.org/abs/2605.09621>.